

Hey, everybody. Welcome to your Chapter 1 review. In this episode, we're going to cover the modern state of health and fitness. We'll dig into some of the more analytical reasons behind the whys of what we do and review key learning points that you'll need to be successful.

Listen, we all know there's a flip side to health and wellness. As fitness pros, we're hyperaware of how important it is to take care of the bodies we've been given. But the truth is, we're seeing more and more cases of obesity driven by increasing automation and sedentary habits. The prevalence of musculoskeletal pain, disease and associated co-morbidities is at an all time high. That's just a fact.

Even without the studies out there, and we'll get into how to sort through all of those in a minute, we just see it in our clients, our friends and family, our kids, and honestly, even in ourselves. So exercise, that's our battle cry, right? Exercise and diet are the keys to good health. OK, yes, but the truth is that, even though diet and exercise can be powerful and necessary tools to improve people's lives, it has to be done properly. If we apply them inappropriately or without a system and science based reasoning, it can contribute to additional pain and injury.

Further, it's critical to understand the importance and complexity of our roles in relation to other Allied Health professionals, including what does and does not fall within our scope of practice as personal trainers. Now, the current global state of health is a complex subject with a lot of moving parts. Information is always changing as science discovers more, and because there are often factors beyond the control of the individual, like genetics, access to health care and preventative medicine, socioeconomics and their environment.

On the other hand, people do have a significant amount of control over their health and well being, but it's all going to come down to their personal decisions to become more physically active and eat healthier foods. This is where you have a tremendous opportunity to influence and empower your clients, to be aware of the things out of their control, and make well-informed decisions about the things they can control. When it comes down to it, evidence-based practice is going to be your best friend. That means using programming or interventions supported by scientific research, derived from previous successes and experiences, and incorporating the preference of your clients.

The good news is NASM has really got your back there. The NASM Optimum Performance Training, or OPT model, provides an ideal framework for evidence-based programming that utilizes a system and exercise program recommendations that are already supported by research, proven to yield real world results and flexible to client preferences. OK, let's get into the nitty gritty of this chapter. You need to be able to distinguish a fitness professional's role within the Allied Health industry, identify the most prevalent chronic health conditions, differentiate evidence based practice from trends and define a CPT scope of practice.

By focusing on scientific principles, NASM systems and methodologies are safe and effective for any client working toward a fitness goal. It's recommended that all fitness professionals also maintain a focus on an evidence-based practice in order to attain the highest levels of success. So how do you separate the proverbial wheat from the chaff in a landscape littered with ideas and articles? Basically, you just have to pay attention to what the scientific research is saying, not whatever the current trends happen to be.

When you strip all the noise away, there are three fundamental components of evidence-based practice, developing individual expertise, staying current on the best sources of external evidence and peer reviewed research, and prioritizing client values and expectations. If you can funnel the programs you design through those three filters, you're going to be in good shape, no pun intended. NASM's OPT model was developed with evidence-based practice as a core guiding philosophy.

The OPT model is flexible and progressive, so it can be applied to any client with any fitness need in nearly any environment. So learn it, take advantage of it, embrace it. It's important to remember that health isn't a static condition. It's ever changing due to the human body's need to adjust to internal and external situations and/or changes in physiology and environments.

Everyone holds a place on the spectrum of health, and sometimes, that's affected by disease. Acute disease is any suddenly occurring condition that can be treated and healed in a short period of time. By contrast, a chronic disease persists without quickly going away or being cured.

Physical inactivity and its relationship to chronic disease is important. A sedentary lifestyle is one of the biggest causes of developing a chronic health condition. Unfortunately, the obvious signs of chronic disease often go unnoticed, until it's too late. However, regular physical activity, as measured by aerobic fitness, is probably the strongest overall predictor of death, disability and disease, and people who maintain their cardiovascular fitness levels across their lifespan are two to four times less likely to develop heart disease or die prematurely.

So you can see why it's so important that we encourage our clients to up their game in terms of physical activity. Let's go over some of the most common concerns. Keeping in mind that you need to be sure your clients are cleared by a medical professional before starting a fitness program, especially if they have any of these conditions. One of the most common challenges facing our current population, obesity.

The terms overweight and obesity refer to a body weight greater than what's considered healthy for a certain height, specifically due to excess body fat. Being overweight or obese greatly increases the chance of developing a chronic disease. Cardiovascular disease is the broad term for any of the numerous problems of the heart and blood vessels, including stroke, heart attacks, heart failure, heart valve problems and arrhythmias. Hypertension is one of the primary risk factors for heart disease and stroke, which are the global leading causes of death.

Cholesterol is a waxy substance found in the blood that is made up of a combination of protein and fatty acids. The human body needs cholesterol, and with a healthy, balanced diet, we produce the amount we need. Chronically high cholesterol can develop fatty deposits in blood vessels and can increase a person's risk for heart disease.

Diabetes, you've undoubtedly heard a lot about this one. Technically speaking, it's a disease that impacts the body's ability to properly metabolize carbohydrates, specifically glucose. With type 1 diabetes, the pancreas doesn't make enough insulin, the hormone that helps transport glucose into cells to provide the energy for work. Without insulin, blood sugar can be raised to dangerous levels, causing numerous health complications.

Type 1 diabetes is typically genetic and unpreventable. However, regular exercise can help people with type 1 diabetes considerably improve their blood glucose management and quality of life. Now, type 2 diabetes is generally linked to obesity. The body still produces insulin, but it's not used properly by the cells. Regulating blood sugar becomes very difficult for these individuals and can also lead to a wide variety of health complications.

Just about everyone knows someone battling cancer or knows a cancer survivor, but what is cancer exactly? Simply put, cancer is an abnormal growth of cells that can result from a wide range of genetic and environmental factors. What is not so simple is that cancer can occur in virtually every part of the body.

There are more than 100 different types of cancer. On the bright side, it's estimated that around 30% to 50% of cancers can be prevented by avoiding risk factors and implementing prevention strategies. Two of those prevention strategies happen to be taking part in regular exercise and eating a healthy diet.

Next, Chronic Obstructive Pulmonary Disease, or COPD, is an umbrella term for lung diseases characterized by increased breathlessness, airflow limitation and accelerated decline of lung function. Though it's not considered curable, it can be medically managed. Lifestyle modification and increased physical activity can have positive effects.

Specifically, light aerobic exercise involving large muscle groups is recommended to help improve the ability to use oxygen. Something, like a brisk daily walk, can be one of the best forms of exercise for people with COPD. Muscular dysfunction should also be taken into consideration. Skeletal muscle development and maintenance is vital to ensure optimal health and well-being across the lifespan of all individuals.

In addition to the increased risk and occurrence of chronic diseases, too much sitting and physical inactivity can cause dysfunction in our body's muscular and skeletal systems. This ultimately increases the risk of both injury and pain in a variety of body regions. At the foot and ankle, two of the most common issues experienced by clients are sprains and plantar fasciitis.

An ankle sprain occurs when a person rolls, twists or turns an ankle, which stretches or tears ligaments. We've all been there. Plantar fasciitis causes pain in the plantar fascia tissue located on the underside of the foot. As the feet and ankles support the entire body, issues there have the potential to impact other areas, such as the knees, hips and/or lower back.

The Lumbopelvic Hip Complex, or LPHC, is made up of the lumbar spine, pelvis, abdomen and hip musculoskeletal structures. It's more commonly referred to as the core. The LPHC is an important anatomical structure, because it connects the upper and lower halves of a person's body. Increasing strength and stability of your core is important for improvements in health, sports performance and prevention of injuries, such as low back pain, which affects up to 80% of adults, at least, once in their lifetime.

Shoulder dysfunction is also very common in the greater population, especially those who frequently lift objects overhead. Because of the shoulders involvement in so many functional movements, it'll be important to keep an eye on shoulder alignment and control with clients of all levels. Headaches and neck pain are becoming more prevalent as well, and are often influenced by altered posture of the head and neck. As with many muscular and skeletal conditions, exercise with emphasis placed on proper alignment and technique may be beneficial in reducing neck discomfort and minimizing injury risk.

Finally, in Chapter 1, we learned about scope of practice as it applies to your role as a CPT and your adjacent Allied Health professionals. All of the relevant local laws and regulations will ensure you're always working within your scope of practice. It's important to understand the complexity of your role, and it is critical that you refer your clients to the appropriate medical professional as soon as you feel you are outside of your scope of practice, period.

NASM established a code of ethics and guidelines as a roadmap to guide CPTs in protecting the public and the profession. Bottom line, follow the NASM Code of Professional Conduct. It's never going to steer you wrong. Learn it like the Pledge of Allegiance, because that's really what it is for us as fitness professionals. It's our pledge to work in the most ethical ways possible.

The need for fitness professionals is at an all time high. There's no doubt that tremendous opportunities exist for you to thrive, while making an impact on society as a CPT. With that, I'm going to end by borrowing a line from the Spider-Man franchise and say that, with great power comes great responsibility. If you take that philosophy seriously, you'll be just fine. Be well, and happy studying.